

Autonomous Gemini SRE Agent & Supportive Reasoning Layer

Target Platform: Google Cloud Platform (GCP)

System of Record: ServiceNow ITSM

Version: 2.0 (Production Blueprint)

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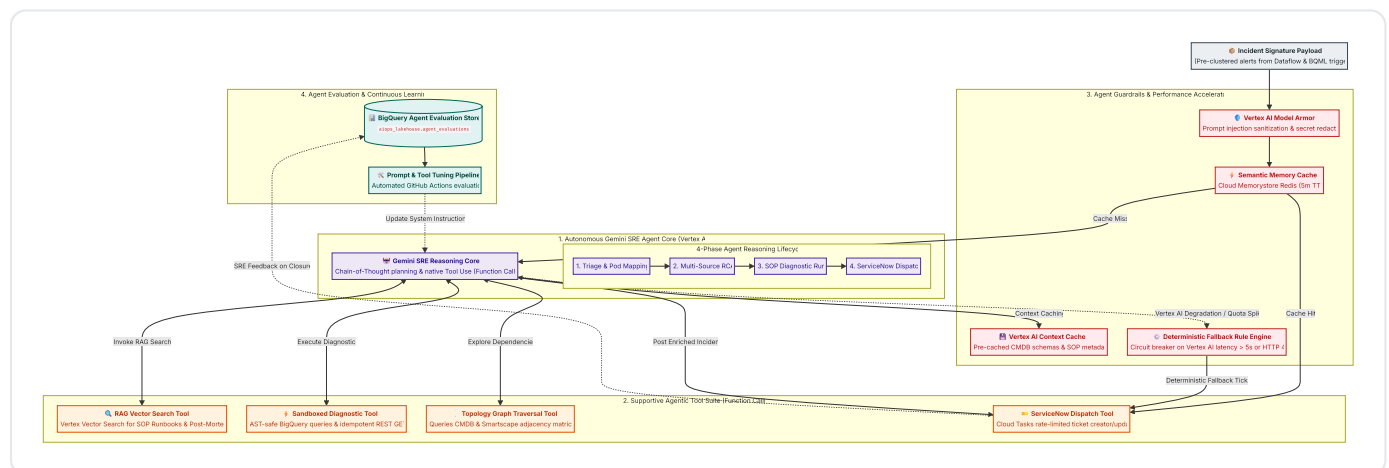
AIops Intelligence & Reasoning Layer - Agent-Centric Architecture

1. Executive Summary & Architectural Role

The **AIops Intelligence & Reasoning Layer** is the cognitive core of the platform, powered natively by **Google Cloud Platform (Vertex AI & Gemini Models)**.

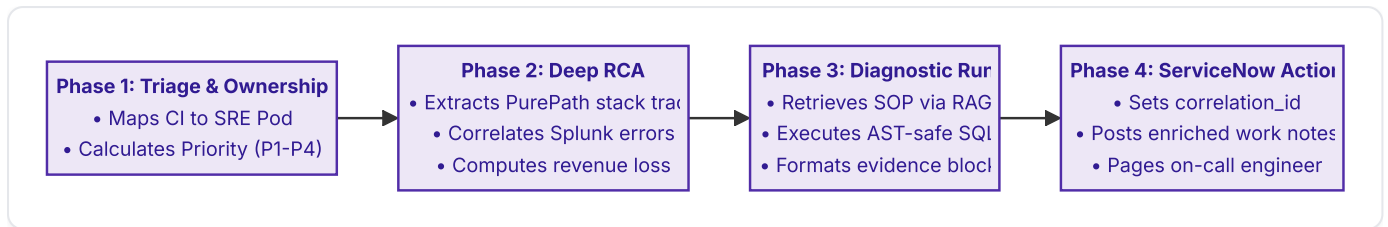
Instead of deploying fragmented LLMs and disconnected execution engines, this layer centers around a unified **Autonomous Gemini SRE Agent** equipped with a specialized **Supportive Agentic Tool Suite**, **Semantic Memory**, **Model Armor Guardrails**, and **Context Caching**.

The agent consumes structured **Incident Signature Payloads** emitted by the [Data Processing & Feature Preparation Layer](#) and autonomously executes a 4-phase reasoning plan to triage incidents, isolate root causes, run non-destructive diagnostics, and update **ServiceNow**.



2. Autonomous Gemini SRE Agent: The 4-Phase Reasoning Lifecycle

Rather than using separate, uncoordinated LLMs for triage and investigation, the **Gemini SRE Agent** executes a single, cohesive reasoning loop via function calling:



Phase 1: Incident Triage & Pod Assignment

- **Objective:** Ingests the structured `Incident Signature Payload` and evaluates impact.
- **Tool Invocation:** Calls `Tool_Topology_Graph` to inspect parent-child microservice dependencies.
- **Outcome:** Assigns urgency, severity, and the responsible SRE pod (e.g., `Checkout-Backend-Pod` vs. `Network-Edge-Pod`), completely bypassing manual L1 helpdesk queues.

Phase 2: Multi-Source Root Cause Analysis (RCA)

- **Objective:** Correlates technical telemetry with business performance.
- **Actions:**
 1. Identifies the failing code path or database session via Dynatrace PurePath trace IDs.
 2. Synthesizes concurrent Splunk server error logs matching the incident time window (`$\pm 10\text{ minutes}`).
 3. Estimates real-time digital financial impact (e.g., *"\$14.2k/min revenue drop based on Adobe Orders Per Minute deviation"*).

Phase 3: RAG Retrieval & Sandboxed Diagnostic Execution

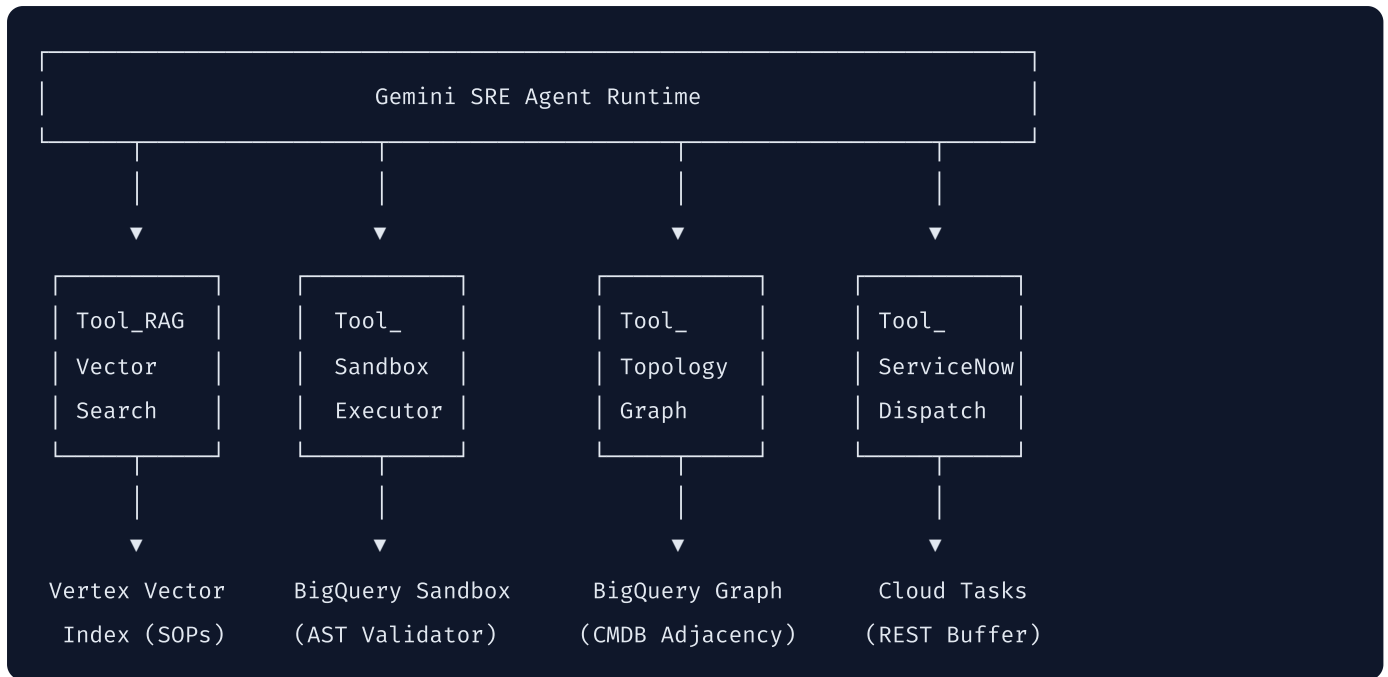
- **Objective:** Autonomous execution of standard diagnostic runbooks before engineer engagement.
- **Actions:**
 1. Calls `Tool_RAG_Search` to retrieve the relevant Standard Operating Procedure (SOP) from **Vertex AI Vector Search**.
 2. Extracts all read-only diagnostic checks from the SOP YAML frontmatter.
 3. Invokes `Tool_Diagnostic_Sandbox` to execute the queries safely against BigQuery and internal monitoring APIs.

Phase 4: ServiceNow Dispatch & Action

- **Objective:** Publish the complete synthesized diagnosis to the enterprise system of record.
- **Actions:**
 1. Constructs the Markdown summary and diagnostic findings block.
 2. Invokes `Tool_ServiceNow_Dispatch` to create or update the incident in ServiceNow with `correlation_id = signature_id`.
 3. Automatically alerts the on-call engineer with complete diagnostic evidence pre-attached.

3. The Supportive Agentic Tool Suite (Function Calling)

The Gemini SRE Agent interacts with the enterprise ecosystem through four strictly governed tools:



3.1 **Tool_RAG_Search** (Runbook & Post-Mortem Retrieval)

- **Underlying Service:** Vertex AI Vector Search Index.
- **Capabilities:** Executes Approximate Nearest Neighbor (ANN) vector search (768-dimension embeddings) filtered by `service_name` and `environment`.
- **Output:** Returns the top-matching Markdown SOP containing symptoms, diagnostic commands, and remediation runbook links.

3.2 **Tool_Diagnostic_Sandbox** (Read-Only Execution Environment)

- **Underlying Service:** BigQuery Client + Cloud Run Gateway with AST Validator.
- **Guardrails:**
 - **AST SQL Parsing:** Blocks mutating statements (`INSERT` , `UPDATE` , `DELETE` , `DROP` , `ALTER` , `TRUNCATE` , `MERGE`). Only `SELECT` queries with mandatory `LIMIT ≤ 100` are permitted.
 - **HTTP Gate:** Restricts external diagnostic API calls strictly to idempotent `GET` requests.
 - **Quotas:** Hard 15-second execution timeout and maximum 1 GB BigQuery scan limit.

3.3 **Tool_Topology_Graph** (Dependency Traversal)

- **Underlying Service:** BigQuery `aiops_lakehouse.topology_service_graph`.
- **Capabilities:** Executes recursive BFS traversal across Dynatrace Smartscape and ServiceNow CMDB CI relationships to return upstream caller and downstream dependency trees.

3.4 **Tool_ServiceNow_Dispatch** (ITSM Enriched Ticket Publisher)

- **Underlying Service:** Asynchronous Cloud Tasks Queue (`aiops-snow-dispatch-queue`).
- **Capabilities:** Submits enriched JSON payloads to ServiceNow with rate-limiting protection (`req/s`) and `correlation_id` deduplication.

4. Agent Guardrails, Security & Performance Acceleration

4.1 Vertex AI Model Armor & Input Sanitization

All raw telemetry strings and external API responses pass through an input sanitizer before entering the Gemini context window:

1. **Prompt Injection Defense:** Strips delimiter attacks (e.g., `Ignore previous instructions`, `SYSTEM PROMPT:`, `Assistant:`).
2. **Secret Redaction:** Regex tokenizers scrub API tokens, private keys, and passwords.
3. **Cloud DLP Integration:** Masks credit card numbers (PCI-DSS) and customer PII from log excerpts.

4.2 Vertex AI Context Caching

- **Mechanism:** Static CMDB microservice schemas, SRE pod matrices, system instructions, and tool definitions are pre-cached in Vertex AI Context Storage.
- **Impact:**
 - **Latency:** Reduces prompt token processing time by up to **80%**.
 - **Cost:** Decreases per-incident input token billing by up to **75%**.

4.3 Semantic Memory Cache (Cloud Memorystore Redis)

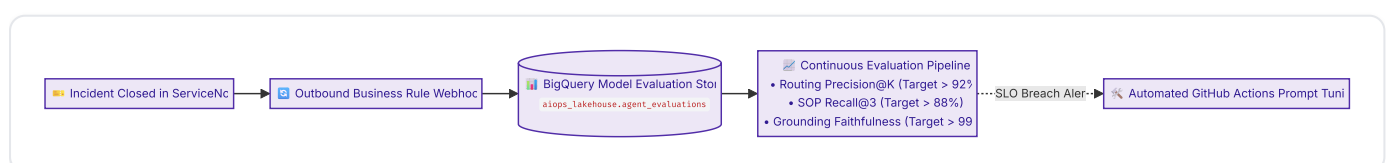
- **Mechanism:** Generates an embedding hash of the `Incident Signature` and queries a 5-minute TTL Redis cache.
- **Benefit:** Reuses RCA and diagnostic findings for recurring or duplicate alert bursts, avoiding redundant LLM invocations during major incidents.

4.4 Deterministic Rule Fallback Engine (Circuit Breaker)

- **Trigger Condition:** If Vertex AI experiences regional degradation, API latency spikes ($> 5\text{ seconds}$), or HTTP 429 quota exhaustion.
- **Action:** The circuit breaker trips immediately. Telemetry is routed through a deterministic Python rule engine using cached CMDB topology tables to assign the ticket, guaranteeing 100% incident routing uptime.

5. Closed-Loop SRE Feedback & Model Observability

To prevent model hallucination drift and continuously improve diagnostic quality, the agent operates in a continuous learning loop:



5.1 Evaluation Metrics & Quality Targets

- **Routing Accuracy (\$> 92%\$):** Percentage of incidents assigned to the correct SRE pod without human re-assignment.
 - **SOP Retrieval Recall@3 (\$> 88%\$):** Percentage of incidents where the relevant remediation runbook was among the top-3 retrieved.
 - **Grounding & Faithfulness (\$> 99%\$):** Verification that all facts, metrics, and trace IDs in the ticket summary are grounded in raw telemetry.
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6. Summary & Cross-Domain Linkages

This agent-centric specification interfaces directly with:

- [Platform Overview & End-to-End Blueprint](#)
 - [Data Processing & Feature Preparation Architecture](#)
 - [Unified Lakehouse Architecture](#)
 - [ServiceNow Integration & Remediation Guide](#)
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